

## 5.1 INTRODUCTION

### Exam Questions Addressed:

- Write the principle of finite difference method. [2079 Shrawan]
- Explain the concept of finite difference method. [2078 Chaitra, 2077 Chaitra, 2073 Bhadra]
- Explain forward, backward and central difference approximations. [2076 Bhadra]
- Describe basic steps in finite difference method. [2075 Bhadra]

## 5.1 Concept & Principle of Finite Difference Method (FDM)

### Concept of FDM:

The fundamental concept of the Finite Difference Method (FDM) is to replace continuous domains (such as space and time) with a discrete grid of points. The continuous partial differential equations (PDEs) are then approximated pointwise by algebraic "difference equations" derived from Taylor series expansions.

### Principle of FDM:

The principle lies in substituting exact, continuous derivatives (which measure instant rates of change) with finite difference quotients over small steps ( $\Delta x, \Delta t$ ). This effectively transforms unsolvable continuous calculus problems into a large system of linear algebraic equations that can be solved numerically by computers.

### Basic Steps in FDM:

1. **Domain Discretization:** Divide the physical domain into a discrete grid in space ( $\Delta x$ ) and time ( $\Delta t$ ).
2. **PDE Discretization:** Replace the governing PDEs with algebraic finite difference equations (FDEs).
3. **Solution Algorithm:** Specify initial/boundary conditions and solve the resulting system of algebraic equations iteratively at each grid point.

### Difference Approximations:

For a function  $f(x)$ , ignoring higher-order truncation errors, the first-order derivatives are approximated as:

#### Forward Difference

$$\frac{\partial f}{\partial x} \approx \frac{f_{i+1} - f_i}{\Delta x}$$

#### Backward Difference

$$\frac{\partial f}{\partial x} \approx \frac{f_i - f_{i-1}}{\Delta x}$$

#### Central Difference

$$\frac{\partial f}{\partial x} \approx \frac{f_{i+1} - f_{i-1}}{2\Delta x}$$



**Exam Questions Addressed:**

- What do you mean explicit and implicit scheme in finite difference method? [2078 Chaitra]
- Explain explicit and implicit finite difference schemes with examples. [2077 Chaitra, 2075 Bhadra]
- What are the differences between explicit and implicit schemes? [2076 Bhadra]
- Discretize using explicit and implicit scheme using forward, backward and central difference. [2073 Bhadra]

## 5.2 Explicit and Implicit Schemes

---

### 1. Explicit Scheme:

In an explicit scheme, the unknown variables at the future time level ( $n + 1$ ) are computed exclusively using known values from the current or past time levels ( $n, n - 1$ ).

- **Advantage:** Very simple to program. Equations are solved sequentially point-by-point.
- **Disadvantage:** Conditionally stable. The time step  $\Delta t$  is strictly limited by the Courant condition to prevent numerical instability.

*Example Expression (Forward Time, Central Space):*

$$u_i^{n+1} = u_i^n + \alpha \frac{\Delta t}{(\Delta x)^2} (u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

### 2. Implicit Scheme:

In an implicit scheme, the unknown variables at the future time level ( $n + 1$ ) are coupled with other *unknown* variables at the same future time level.

- **Advantage:** Generally unconditionally stable. Allows for much larger computational time steps ( $\Delta t$ ).
- **Disadvantage:** Mathematically complicated and computationally heavy. Requires solving a system of simultaneous equations at every step (e.g., using Newton-Raphson or matrix inversion).

*Example Expression (Backward Time, Central Space):*

$$u_i^{n+1} = u_i^n + \alpha \frac{\Delta t}{(\Delta x)^2} (u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1})$$

**Spatial Discretization Expressions for  $\frac{\partial f}{\partial x}$ :**

### Explicit FDEs (Evaluated at time $n$ )

---

Forward:  $\frac{f_{i+1}^n - f_i^n}{\Delta x}$

Backward:  $\frac{f_i^n - f_{i-1}^n}{\Delta x}$

Central:  $\frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x}$

### Implicit FDEs (Evaluated at time $n + 1$ )

---

Forward:  $\frac{f_{i+1}^{n+1} - f_i^{n+1}}{\Delta x}$

Backward:  $\frac{f_i^{n+1} - f_{i-1}^{n+1}}{\Delta x}$

Central:  $\frac{f_{i+1}^{n+1} - f_{i-1}^{n+1}}{2\Delta x}$



**Exam Questions Addressed:**

- Write down the governing equations describing movement of fluid. [2078 Chaitra, 2078 Kartik, 2074 Magh, 2071 Magh, 2070 Bhadra]
- What are the Kinematic wave approximations of these governing equations? [Multiple Years]
- Describe first-order and second-order schemes with graphs. [2079 Jestha, 2073 Magh]

## 5.3 Governing Equations & Kinematic Approximation

The movement of fluid in open channels is governed by the 1D Saint-Venant Equations, derived from the conservation of mass and momentum:

### 1. Continuity Equation (Conservation of Mass):

$$\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_L$$

### 2. Momentum Equation (Conservation of Momentum):

$$\underbrace{\frac{1}{g} \frac{\partial V}{\partial t}}_{\text{Local Accel.}} + \underbrace{\frac{V}{g} \frac{\partial V}{\partial x}}_{\text{Convective Accel.}} + \underbrace{\frac{\partial y}{\partial x}}_{\text{Pressure Gradient}} = \underbrace{S_0 - S_f}_{\text{Gravity \& Friction}}$$

### Kinematic Wave Approximation:

In steep channels or slow-rising floods, the local acceleration, convective acceleration, and pressure gradient terms are negligibly small compared to the bed slope and friction. By dropping them, we get the **Kinematic Wave Approximation**:

$$S_0 = S_f$$

This implies that the gravitational force perfectly balances the frictional force, meaning discharge  $Q$  is strictly a function of flow area  $A$  (e.g.,  $Q = \alpha A^m$ ).

## First-Order and Second-Order Schemes

### 1. First-Order Accurate Scheme (e.g., FTBS Explicit)

A first-order scheme uses a two-point difference approximation (forward or backward) for the derivatives, resulting in a truncation error of  $O(\Delta x, \Delta t)$ .

$$\text{Forward Time: } \frac{\partial f}{\partial t} \approx \frac{f_i^{n+1} - f_i^n}{\Delta t}, \quad \text{Backward Space: } \frac{\partial f}{\partial x} \approx \frac{f_i^n - f_{i-1}^n}{\Delta x}$$

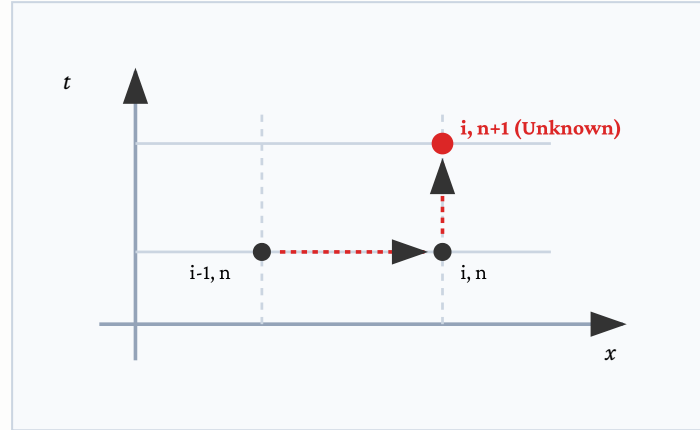


Figure: First-Order Explicit Stencil (Forward Time, Backward Space)

## 2. Second-Order Accurate Scheme (e.g., CTCS Leap-Frog Explicit)

A second-order scheme uses a three-point central difference approximation, yielding a much smaller truncation error of  $O(\Delta x^2, \Delta t^2)$ .

$$\text{Central Time: } \frac{\partial f}{\partial t} \approx \frac{f_i^{n+1} - f_i^{n-1}}{2\Delta t}, \quad \text{Central Space: } \frac{\partial f}{\partial x} \approx \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x}$$

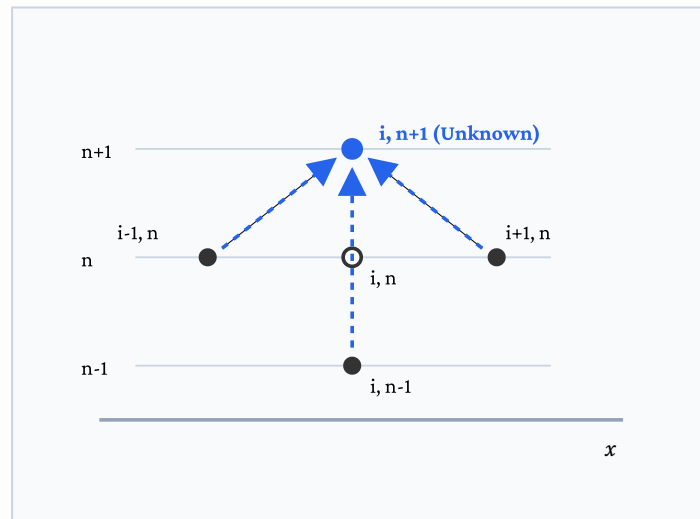


Figure: Second-Order Leap-Frog Explicit Stencil (CTCS)



### Exam Questions Addressed:

- Derive the kinematic wave approximation for movement of fluid. [2070 Bhadra]
- Derive 1st-order accurate implicit FDE for linear kinematic wave + parameters of momentum eq. [2080 Chaitra]
- Write an algorithm for the 2nd order accurate solution of kinematic wave. [2071 Magh]
- Derive a 2nd order accurate finite difference scheme of linear kinematic wave. [2070 Bhadra]

## 5.4.1 Linear Kinematic Wave Model

By taking the kinematic approximation ( $S_0 = S_f$ ), we find  $Q = \alpha A^m$ . Differentiating w.r.t. time yields  $\frac{\partial A}{\partial t} = \frac{1}{c} \frac{\partial Q}{\partial t}$ , where  $c = \frac{dQ}{dA}$  is the wave celerity. Substituting this into the continuity equation gives the **Linear Kinematic Wave Equation**:

$$\frac{\partial Q}{\partial t} + c \frac{\partial Q}{\partial x} = 0$$

### 1. First-Order Accurate Implicit Scheme (Backward Space, Backward Time):

Evaluating the spatial derivative at the unknown future time level  $n + 1$  guarantees an unconditionally stable implicit formulation:

$$\begin{aligned} \frac{Q_i^{n+1} - Q_i^n}{\Delta t} + c \left( \frac{Q_i^{n+1} - Q_{i-1}^{n+1}}{\Delta x} \right) &= 0 \\ Q_i^{n+1} - Q_i^n + C_r(Q_i^{n+1} - Q_{i-1}^{n+1}) &= 0 \quad (\text{where } C_r = c\Delta t/\Delta x) \\ Q_i^{n+1}(1 + C_r) &= Q_i^n + C_r Q_{i-1}^{n+1} \\ Q_i^{n+1} &= \frac{Q_i^n + C_r Q_{i-1}^{n+1}}{1 + C_r} \end{aligned}$$

### 2. Second-Order Accurate Scheme (Central Time, Central Space):

Using central differences gives a second-order accurate formulation computing discharge for unknown time and location:

$$\begin{aligned} \frac{Q_i^{n+1} - Q_i^{n-1}}{2\Delta t} + c \left( \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} \right) &= 0 \\ Q_i^{n+1} &= Q_i^{n-1} - C_r(Q_{i+1}^n - Q_{i-1}^n) \end{aligned}$$

### 3. Parameters of the Momentum Equation:

For a wide rectangular channel,  $R \approx y$  and Area  $A = by$ . From Manning's equation:

$$Q = \frac{1}{n} AR^{2/3} S_0^{1/2} = \frac{1}{n} A \left( \frac{A}{b} \right)^{2/3} \sqrt{S_0} = \left( \frac{\sqrt{S_0}}{nb^{2/3}} \right) A^{5/3}$$

Comparing this to the general kinematic momentum equation  $Q = \alpha A^m$ , the parameters are:

$$\alpha = \frac{\sqrt{S_0}}{nb^{2/3}} \quad \text{and} \quad m = \frac{5}{3}$$

### 4. Algorithm for Second-Order Accurate Solution:

1. Supply initial conditions  $Q_i^0$  at time  $t = 0$  for all spatial points  $i$ .
2. Supply upstream boundary conditions  $Q_1^n$  for all time steps.
3. Advance time step  $t = t + \Delta t$ .
4. For interior nodes ( $i = 2$  to  $N - 1$ ), compute  $Q_i^{n+1}$  using the second-order CTCS equation:  
$$Q_i^{n+1} = Q_i^{n-1} - C_r(Q_{i+1}^n - Q_{i-1}^n).$$
5. For the downstream boundary node ( $N$ ), use a first-order backward space approximation to compute the unknown boundary.
6. Repeat steps 3 to 5 until the final simulation time is reached.



**Exam Questions Addressed:**

- Derive the space-time discretization of a second-order non-linear kinematic wave. [2079 Shrawan]
- Derive the 1st order accurate implicit FDE for kinematic wave model in non-linear form. [2071 Bhadra]
- Define Courant number and condition. [2078 Kartik, 2077 Chaitra]
- Describe numerical dispersion, diffusion, and stability. [2079 Jestha, 2074 Bhadra, 2077 Chaitra]

## 5.4.2 Non-Linear Scheme for Kinematic Wave

### 1. First-Order Accurate Implicit Scheme (Non-Linear):

The non-linear continuity equation is  $\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q_L$ . Using a first-order fully implicit backward-time, backward-space scheme:

$$\frac{A_{i+1}^{n+1} - A_{i+1}^n}{\Delta t} + \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} = q_{L,i+1}^{n+1}$$

Substituting the non-linear area-discharge relationship  $A = \alpha Q^\beta$  evaluated at the future time step:

$$\frac{\alpha(Q_{i+1}^{n+1})^\beta - \alpha(Q_{i+1}^n)^\beta}{\Delta t} + \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} = q_{L,i+1}^{n+1}$$

This produces a non-linear algebraic equation for  $Q_{i+1}^{n+1}$  that must be solved using Newton-Raphson iteration.

### 2. Second-Order Accurate Implicit Scheme (Non-Linear):

To achieve second-order accuracy in space and time, the Preissmann 4-point implicit scheme is used with a weighting factor  $\theta = 0.5$ :

$$\frac{A_{i+1}^{n+1} + A_i^{n+1} - A_{i+1}^n - A_i^n}{2\Delta t} + \frac{Q_{i+1}^{n+1} - Q_i^{n+1}}{\Delta x} = \bar{q}_L$$

### 5.4.3 & 5.4.4 Stability, Diffusion, and Dispersion

---

#### Stability & Courant Condition:

**Stability** ensures that numerical errors (round-off and truncation) do not grow exponentially during the simulation.

**Courant Number ( $C_r$ )** is defined as the ratio of the physical wave celerity ( $c$ ) to the numerical grid speed ( $\Delta x / \Delta t$ ).

$$C_r = \frac{c\Delta t}{\Delta x}$$

**Courant Condition** requires that for explicit schemes to remain stable, the Courant number must be less than or equal to 1 ( $C_r \leq 1$ ).

#### Numerical Diffusion & Dispersion:

- **Numerical Diffusion:** Caused by even-order truncation errors (e.g.,  $\partial^2 / \partial x^2$ ) inherent in the difference approximation. It artificially "smears", flattens, or attenuates the peak of a passing wave, simulating an artificial physical friction.
- **Numerical Dispersion:** Caused by odd-order truncation errors (e.g.,  $\partial^3 / \partial x^3$ ). It forces different frequency components of the numerical wave to travel at slightly different speeds, resulting in artificial "wiggles" or oscillations trailing or leading the main wave front.



## 5.5 DYNAMIC WAVE MODEL (SAINT-VENANT SOLVERS)

### Exam Questions Addressed:

- Derive the expression for Leap frog explicit finite equation for dynamic wave model. [2079 Chaitra]
- Derive FDE for full Saint-Venant equations using 2nd order accurate explicit scheme. [2074 Bhadra, 2072 Ashwin]
- Discretize the Saint Venant equations using implicit four point method. [2072 Magh]

## 5.5 Solutions to Dynamic Wave Model

### 1. Solution by Explicit Scheme (Leap-Frog Method)

The Leap-Frog scheme utilizes Central-Time, Central-Space (CTCS) approximations. It is second-order accurate in both space and time,  $O(\Delta x^2, \Delta t^2)$ .

$$\frac{\partial f}{\partial t} \approx \frac{f_i^{n+1} - f_i^{n-1}}{2\Delta t}$$
$$\frac{\partial f}{\partial x} \approx \frac{f_{i+1}^n - f_{i-1}^n}{2\Delta x}$$

Substituting into the full Saint-Venant continuity equation ( $\frac{\partial A}{\partial t} + \frac{\partial Q}{\partial x} = q$ ):

$$\frac{A_i^{n+1} - A_i^{n-1}}{2\Delta t} + \frac{Q_{i+1}^n - Q_{i-1}^n}{2\Delta x} = q_i^n$$

Solving explicitly for the future area  $A_i^{n+1}$ :

$$A_i^{n+1} = A_i^{n-1} - \frac{\Delta t}{\Delta x} (Q_{i+1}^n - Q_{i-1}^n) + 2\Delta t q_i^n$$

Similarly, substituting CTCS into the momentum equation (

$\frac{\partial Q}{\partial t} + \frac{\partial(Q^2/A)}{\partial x} + gA \frac{\partial y}{\partial x} - gA(S_0 - S_f) = 0$ ) allows explicit isolation of  $Q_i^{n+1}$ .

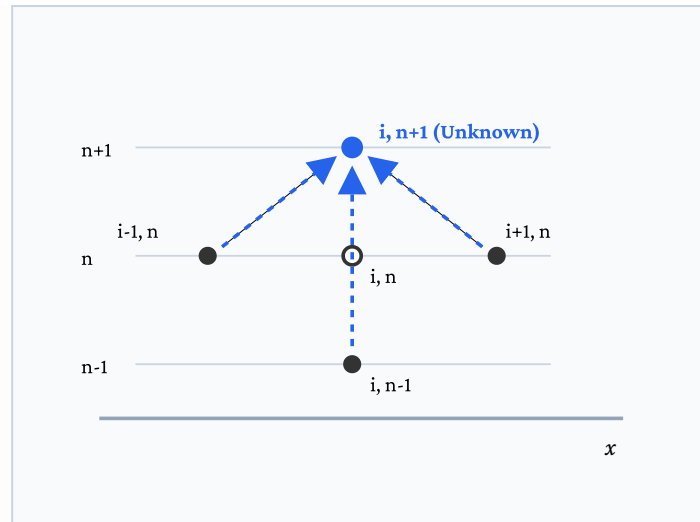


Figure: Second-Order Leap-Frog Explicit Stencil (CTCS)

## 2. Solution by Implicit Four-Point Method (Preissmann Scheme)

This scheme evaluates spatial derivatives, time derivatives, and function values at an interior point  $M$  defined by a temporal weighting factor  $\theta$  (typically  $0.5 \leq \theta \leq 1.0$ ) and a spatial weighting factor of 0.5.

$$f(x, t) \approx \frac{\theta}{2}(f_{i+1}^{n+1} + f_i^{n+1}) + \frac{1-\theta}{2}(f_{i+1}^n + f_i^n)$$
$$\frac{\partial f}{\partial t} \approx \frac{f_{i+1}^{n+1} + f_i^{n+1} - f_{i+1}^n - f_i^n}{2\Delta t}$$
$$\frac{\partial f}{\partial x} \approx \theta \left[ \frac{f_{i+1}^{n+1} - f_i^{n+1}}{\Delta x} \right] + (1-\theta) \left[ \frac{f_{i+1}^n - f_i^n}{\Delta x} \right]$$

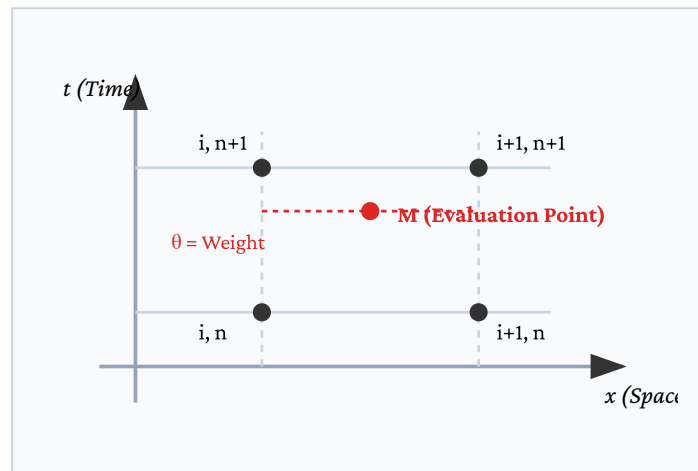


Figure: Preissmann Implicit Four-Point Stencil

Substituting these approximations into the non-linear Saint-Venant equations generates a large system of simultaneous algebraic equations, which is unconditionally stable for  $\theta \geq 0.5$ .